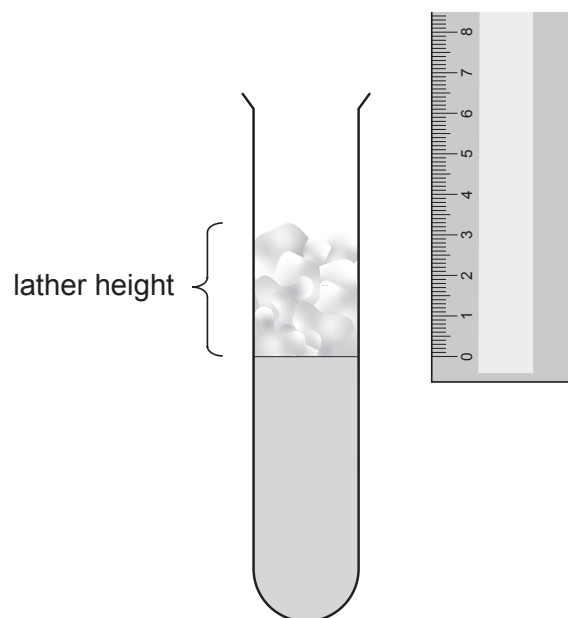


5. (a) A class measured the hardness of five different samples of water, **A**, **B**, **C**, **D** and **E**.

They added soap solution to each water sample and measured the height of the lather produced on shaking.



Their results are shown in the table.

Water sample	Lather height (cm)
A	1.5
B	3.0
C	0.5
D	5.0
E	2.0



- (i) Complete the order of hardness of the water samples, from softest to hardest. [1]

softest $\longrightarrow$ hardest

D				C
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- (ii) Calcium is one of two metal ions which cause hardness in water.

Underline the name of the other metal ion which causes hardness in water. [1]

potassium sodium nickel magnesium tin

- (iii) Tick (✓) **all** of the variables which must be controlled in this experiment. [2]

volume of soap solution

☐

type of water

☐

type of soap solution

☐

volume of water

☐

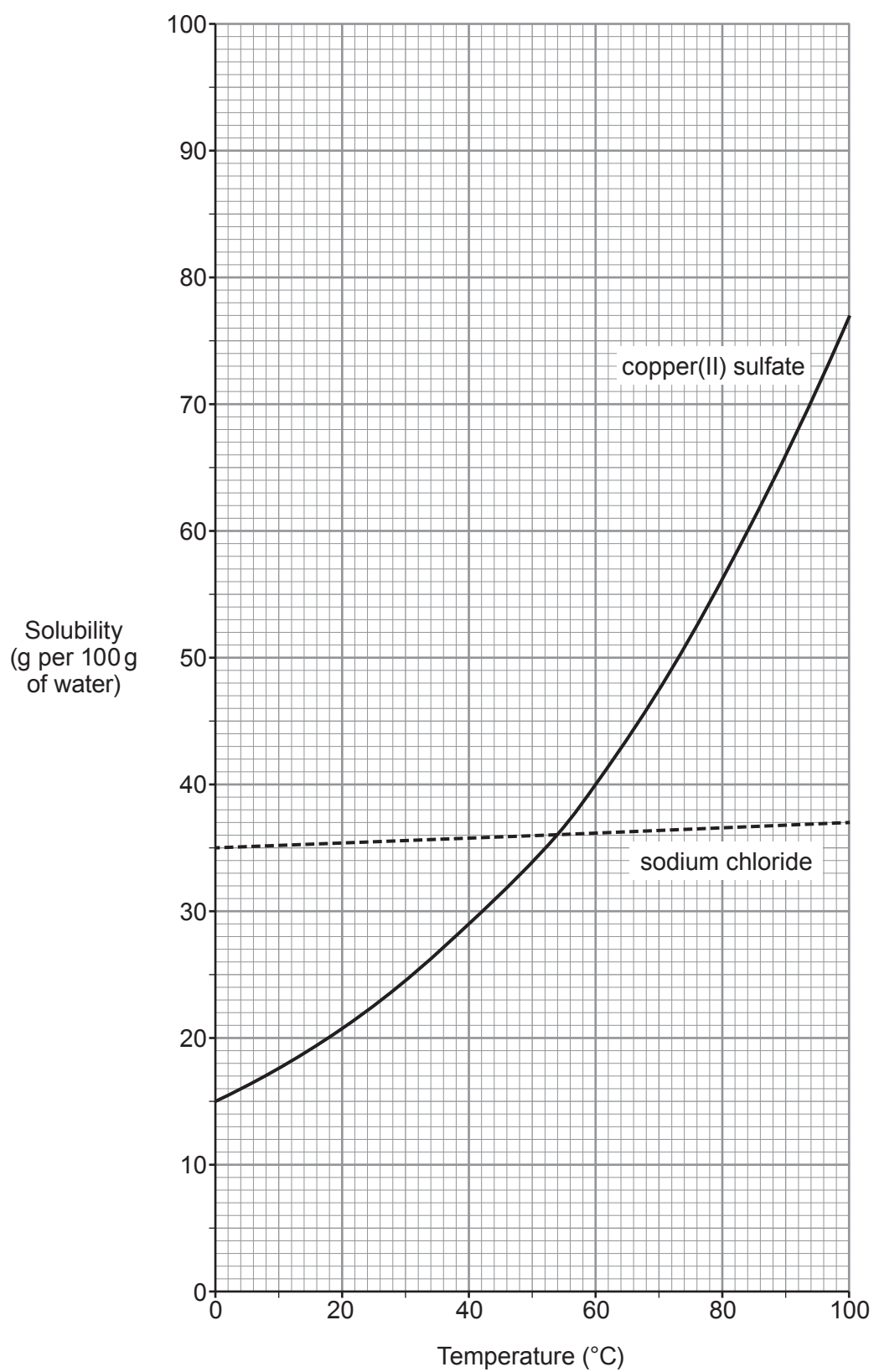
height of lather

☐

width of test tube

☐

- (b) The graph shows the solubilities of sodium chloride and copper(II) sulfate at different temperatures.



- (i) State the temperature at which the two compounds have the same solubility. [1]
 °C

- (ii) Compare the changes in solubility of the two compounds. [2]

.....

.....

.....

- (iii) Calculate the mass of copper(II) sulfate that would dissolve in **1000 g** of water at 40 °C. [2]

Mass = g

- (c) Potassium sulfate, K_2SO_4 , is soluble in water.

- (i) State the number of sulfur atoms in the formula K_2SO_4 . [1]

.....

- (ii) Calculate the relative formula mass (M_r) of K_2SO_4 . [2]

$$A_r(K) = 39 \quad A_r(S) = 32 \quad A_r(O) = 16$$

$M_r =$

